

# Replication Instructions for: Rising Markups, Rising Prices?

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Conlon, Miller, Otgon, Yao (2023)

A copy of the paper is here: [https://github.com/TsolmonO/Rising\\_markups](https://github.com/TsolmonO/Rising_markups)

## Open Instructions

1. Download and unzip the repository
2. All required files are included or are downloaded programmatically from WRDS, BLS, and FRED.
3. "RUN\_ALL.py" file in the "Code" folder automatically downloads the data and generates the output.  
Please read the details about the data downloads below before running the code.

## Dataset Size and Memory

1. All datasets saved will take up about 440MB of drive space and takes about 3 minutes.
2. Total runtime on Windows 11 with 40GB of RAM is approximately 5 minutes.
3. All files and output will take up about 895MB of drive space.

## Data downloads

### Downloading data from WRDS

There are two options for downloading data from WRDS

1. Download programmatically using Web Query:

The main "RUN\_ALL.py" file will call "0.0 Download Compustat.py" and the user will be prompted for username and password.

To request an account please visit: <https://wrds-www.wharton.upenn.edu/>

After entering the username and password, we suggest creating a pgpass file that stores your credentials.

In addition, you may need to downgrade your SQLAlchemy to 1.4.x as the recent upgrade does not support the backwards compatibility.

2. Download manually from WRDS website. After accessing Compustat using your username and password choose the following: (These instructions follow replication files from De Loecker, Eeckhout, Unger (2020))

De Loecker, Jan; Eeckhout, Jan; Unger, Gabriel, 2020, "Replication Data for: 'The Rise of Market Power and the Macroeconomic Implications' ", <https://doi.org/10.7910/DVN/5GH8XO>, Harvard Dataverse, V1

1. North America
2. Fundamentals Annual (Fundamental Quarterly for quarterly data)
3. Date range: 1955 - 2021 (2018Q1 - 2022Q3 for quarterly)

4. GVKEY code: search the entire database
5. Consolidated accounts, format INDL and FS (drop FS if reported both), domestic population source, data format STD
6. Data selected:
  - Annual data: gvkey, indfmt, consol, popsrc, datafmt, conm, fyear, sale, cogs, xsga, ppegt.
  - Quarterly data: gvkey, indfmt, consol, popsrc, datafmt, conm, fyearq, fqtr, saleq, cogsq, xsgaq, ppegtq.
7. Save files as Stata files named "Compustat\_annual.dta" and "Compustat\_quarterly.dta" in the directory "Input/DLEU".

## Downloading data from BLS

Our replication files automatically download Producer Price Index data from

<https://download.bls.gov/pub/time.series/pc/>. If you wish to download the data manually, please access this directory and download the files starting with "pc.data".

## Downloading data from FRED:

If you wish to download the CPI data programmatically, please enter your API key in "path\_plot\_config.py" file located in "Code" folder. Alternatively, you can download the data from here:

<https://fred.stlouisfed.org/series/CPIAUCSL>

## Software dependencies

### Stata location

Markups are calculated using Stata code based on De Loecker, Eeckhout, Unger (2020) replication files. Please make sure you have Stata installed and provide the path to your Stata directory in Step 1 in "RUN\_ALL.py" and Line 10 in "0.3 Create Main Datasets.do" files.

### Python dependencies

Our replication package uses several Python packages. Please see the list of necessary packages in "requirements.txt" located in the main directory.

To install these dependencies (such as on a shared server) you can run the following:

Python (version 3.8 or above) - install dependencies with

```
pip3 install -r requirements.txt
```

## How to run code

Run "RUN\_ALL.py" file located in "Code" folder to run all files.

## File of origin for tables and figures

| Table/Figure                       | Generating File                                   |
|------------------------------------|---------------------------------------------------|
| Figure 1                           | Code/1. Generate Figure 1 - Aggregate Markup      |
| Figure 2                           | Code/2. Generate Figure 2 - CAGR of PPI vs Markup |
| Figure 2 for 2-digit industries    | Code/2. Generate Figure 2 - CAGR of PPI vs Markup |
| Outliers (Annual and Quarterly)    | Code/2. Generate Figure 2 - CAGR of PPI vs Markup |
| Stats Table (Annual and Quarterly) | Code/3. Generate Summary Statistics               |
| Table 1 (CAGR WLS(COGS))           | Code/4. Generate Table 1                          |

## Files Provided and Data Access Statements

### WRDS

We use Compustat data from WRDS. The data is downloaded programmatically through WRDS API or manually (instructions provided above) and we are not able to include individual files in this replication package. These include: Fundamentals Annual and Fundamentals Quarterly (See terms: <https://wrds-www.wharton.upenn.edu/users/tou/>)

### Publicly available data

#### BLS

The Producer Price Index data is downloaded programmatically from BLS website using our files and we did not include the data in our replications package. Please see Data Downloads section above for details.

#### FRED

The Consumer Price Index data is downloaded programmatically from FRED website using our files and we did not include the data in our replications package. Please see Data Downloads section above for details.

### Authors constructed files

- "Input/Other/indexed\_quarter.xlsx" contains numeric indices for quarters analyzed.
- "NAICS\_2D\_Description.xlsx" contains descriptions for 2-digit NAICS codes.

### Files from other sources

- The original file for "Input/DLEU/macro\_vars\_new.xlsx" is provided in De Loecker, Eeckhout, Unger (2020) replication files. We have updated the data using the publicly available FRED data until 2022.
- The original file for "Input/DLEU/PF/theta\_W\_s\_window\_new.do" is provided in De Loecker, Eeckhout, Unger (2020) replication files.
- "Intermediate/Aggregate Markups by DLEU.xlsx": markups from Figure 1 of De Loecker, Eeckhout, Unger (2020).